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Marketplace Chameleons: Demystifying the Scam Ecosystem of Second-Hand Online Platforms

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Abstract*—Second-hand online marketplaces have become a major channel for consumer-to-consumer commerce, but their low-friction and weakly mediated transaction environment also creates fertile ground for scams. Despite the growing impact, existing studies and regulatory efforts largely focus on individual scam types, leaving the broader ecosystem and scam organization poorly understood. In this paper, we present the first ecosystem-level measurement study of scams on second-hand online marketplaces by linking downstream victim reports to upstream scammer-side signals. On the downstream side, we collected victim-reported narratives from online communities and applied a large language model (LLM)-assisted pipeline for large-scale analysis and iterative taxonomy construction. This process yielded 3,392 scam cases across four major platforms (i.e., eBay, Craigslist, Facebook Marketplace, and Goofish) and resulted in a 43-category taxonomy. Our analysis shows that the structure of second-hand marketplaces creates a diverse scam ecosystem, where scammers operate as both buyers and sellers and target all stages of the transaction. Looking into the cases, scammers focus on high-liquidity products and employ psychological tactics to manipulate victims, with an average reported loss of $841.03 per incident. In addition, these scams are influenced by platform design. Stronger protections, such as receipt confirmation and refund policies, often reshape rather than prevent scams. On the upstream side, we traced 88 scammer-authored posts to uncover upstream organizational signals and identified a proxy-driven scam model from public Telegram channels. The low entry barriers of second-hand marketplaces enable organizers to recruit proxies at scale, suggesting that upstream signals offer a proactive lens for early intervention. Overall, we present a systematic view of scams on second-hand marketplaces and their proxy organization, offering insights for more effective detection and defense.

I. Introduction

Second-hand online marketplaces have rapidly expanded the scale and diversity of consumer-to-consumer (C2C) commerce [1], enabling low-friction trading of heterogeneous goods among unvetted participants. Unlike business-to-consumer (B2C) platforms, second-hand marketplaces commonly feature weaker platform mediation, more flexible payment and delivery arrangements, and greater ambiguity in transaction roles. These properties make the ecosystem attractive to adversaries who impersonate buyers or sellers, manipulate payment and shipping procedures, and exploit victims' trust to obtain money, goods, or sensitive information. As widely reported [2], [3], [4], such scams pose persistent challenges for users and platform governance.

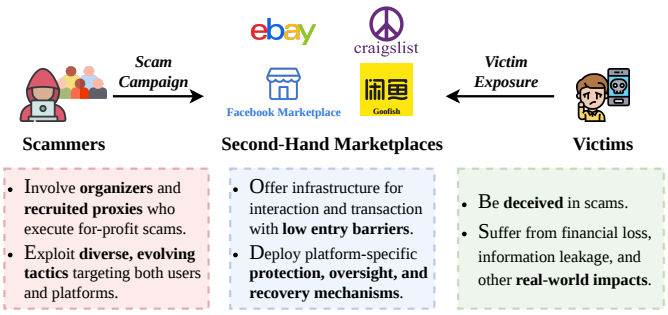


Fig. 1: Scam ecosystem of second-hand online platforms.

Despite their growing practical impact, our understanding of scams on second-hand marketplaces remains narrowly incident-centered. Existing research [5], [6], [7] and regulatory efforts [8], [9] largely examine individual scam types in isolation, focusing on specific fraud archetypes through a microscope and often drawing evidence from general e-commerce. While such work characterizes fraud mechanisms and informs targeted mitigation, it rarely considers scams on second-hand C2C marketplaces from an ecosystem perspective that encompasses diverse scam categories, coordinated actors, and structured campaigns. As a result, how these scams are organized and operate at scale remains poorly understood.

Our Work. In this work, we present the first systematic measurement of scams on second-hand online marketplaces, referred to as SHScams, moving beyond incident-level analysis to an ecosystem-level view. We study four major C2C platforms, namely eBay, Facebook Marketplace, Craigslist, and Goofish, which together cover a broad user base [10], [11], [12], [13]. As shown in Figure 1, our study centers on three key stakeholders: scammers, victims, and marketplaces, and aims to ask three key questions shaped by the unique characteristics of second-hand markets:

• *RQ1*: How diverse are SHScams in the roles that scammers assume, and what are the impacts of these scams on victims?

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• *RQ2*: How do scams adapt to platform-specific regulations and evolve under these constraints?
• *RQ3*: Are SHScams organized beyond isolated incidents, and if so, what patterns can be observed?

To answer these questions, we construct a scam dataset from scam-related posts on public online communities. Given the limited accessibility of platform transaction logs and officially confirmed scam records, these posts provide one of the few available sources of detailed information on scam execution, victim impact, and occasional upstream organizer activity. Although these self-reported data may be subject to reporting and selection biases, analyzing them at scale allows us to identify recurring scam patterns beyond isolated incidents, examine their diversity and evolution, and uncover the organizational structures underlying them. We therefore collected 11,701 scam-related posts from Reddit, Douban, Zhihu, and RedNote, and conducted a large-scale systematic analysis.

We use this dataset to study the diversity of SHScams (RQ1) and their adaptation to marketplace constraints (RQ2). Due to the heterogeneity of forum content [14], [15], we first apply semantic filtering to select relevant posts. Our primary analysis then focuses on detailed victim-reported experiences to capture these scam patterns systematically. A key challenge is the diversity and rapid evolution of SHScams, which makes each case difficult to characterize without deep semantic understanding. To address this challenge, we develop an LLM-assisted processing pipeline that iteratively constructs a taxonomy tailored to second-hand marketplaces. Each retained report is converted into a structured case capturing roles, traded items, behavioral sequences, and outcomes, before being assigned to a scam category. This process reveals 43 scam categories across 3,392 cases.

We further use a subset of this dataset to analyze scam organization (RQ3). Specifically, we identify 88 scammer-authored posts that can be linked not only to second-hand marketplace activities but also to upstream recruitment efforts that precede the corresponding SHScams. To connect these fragmented signals, we conduct an infiltration analysis by interacting with recruiters on Telegram and gathering insights into the organizational structure of SHScams. To our knowledge, this is the first study to link upstream Telegram-based recruitment with downstream victim reports from online communities, revealing the distinctive organizational properties of second-hand marketplace scams.

Main Findings. In our study, we show that the scam ecosystem on second-hand marketplaces spans 43 categories with varied execution patterns, far beyond a few familiar fraud types (see §IV). This diversity is driven by role ambiguity and flexible transaction workflows, enabling scammers to act as sellers, buyers, or both, target different transaction stages, and extend beyond transaction fraud to external deception such as task scams. Scammers target high-liquidity products and rely on psychological manipulation, such as threats, to secure victim compliance. Based on self-reported cases, the average financial loss per incident is $841.03, with *Buyer-Induced Seller Payments* (Group B) scams accounting for the highest average loss per incident. In addition, scam distributions differ substantially across platforms, shaped by platform positioning, user populations, and platform policies (see §V). Scammers move interactions and transactions off-platform to evade platform oversight and adapt to platform-specific protections, shifting to scam patterns that mimic legitimate behavior or exploit protection workflows. While overall recovery is difficult, platforms like Goofish (39.05%) and eBay (36.48%) achieve higher recovery rates, highlighting the key role of platform mediation in recovery outcomes.

Our upstream analysis further shows that some scams operate as coordinated campaigns rather than isolated incidents (see §VI). Using insights from scammer-authored posts, we examined 24 public Telegram channels and identified a proxy-driven scam model that decouples recruitment from execution to evade platform detection. Our validation further uncovered 235 potential real-world scam listings using 90 images sourced from these channels, enabling early detection of scam activity across marketplaces and supporting platform governance.

Contributions. Our contributions are as follows:
• *Ecosystem-level measurement*. We present the first systematic study of SHScams, analyzing 3,392 cases across four major C2C platforms. This work provides an ecosystem-level perspective that captures scam categories, cross-platform behaviors, and structured campaigns, beyond individual cases.
• *Scam characterization*. Using an LLM-assisted approach, we analyze victim-reported data to examine how scams are executed and their impacts on victims. We construct a 43-category taxonomy capturing evolving scams, revealing a diverse ecosystem shaped by second-hand marketplaces.
• *Campaign discovery*. We identify a proxy-driven scam model in which upstream organizers recruit proxies to carry out specific scams. These upstream recruitment signals correspond to several downstream scam categories and can serve as early indicators for fraud detection.

II. Background

Second-hand trading has become widely adopted [16], driving the rapid growth of online second-hand marketplaces. In this section, we introduce the representative platforms studied in this paper, and formalize the threat model to capture the scam scenarios occurring on these marketplaces.

A. Popular Second-Hand Marketplaces

Second-hand marketplaces are online platforms that facilitate the exchange of goods between individual users. While all of them support C2C transactions, they differ substantially in communication control, payment mediation, and transaction protection. These differences shape the opportunities available to scammers. To capture this variation, we focus on four representative platforms with distinct design characteristics.
• *eBay* [17], founded in 1995, is a major online marketplace with approximately 132 million active users as of 2024 [10], [18]. The platform supports buyer-seller communication through internal messaging and also permits off-platform contact through email or phone. eBay officially

TABLE I: Popular second-hand marketplaces and details.

Marketplace Name	Target Area	Protection Mechanism	Internal Messaging	Online Payment
eBay	Global	Yes	Yes	Yes
Craigslist	Global	No	No	No
Facebook Marketplace	Global	No	Yes	Yes
Goofish	China	Yes	Yes	Yes

recommends online payment methods (e.g., PayPal) to enhance transaction security. It also provides customer protection through its "Money Back Guarantee" [19], which enables buyers to request a refund if an item does not meet expectations or if the transaction fails to be completed as promised.

• *Craigslist* [20], launched in 1995, is the most visited classified listings platform in the United States [11]. It is widely used for second-hand goods transactions and recorded 112.4 million monthly visits in October 2025 [21]. The platform operates with minimal intervention, allowing sellers to publicly share contact information for direct buyer–seller communication. It does not support integrated payments or provide transaction protection, resulting in a highly decentralized environment where local, offline interactions are prevalent.

• *Facebook Marketplace* [22], introduced in 2016, is a second-hand trading platform integrated within Facebook. It encourages on-platform communication via Facebook Messenger while supporting both online and offline transactions. The platform does not provide customer protection or formal dispute resolution mechanisms. However, its integration within Facebook has driven widespread adoption, with over one billion active users across multiple regions each month [12].

• *Goofish* [23], launched by Alibaba in 2014, is a leading second-hand marketplace in China with approximately 162 million monthly active users [13]. The platform provides an integrated messaging system and requires online payments via Alipay. The platform provides customer protection through an escrow-based payment model [24], [25], where funds are held by the platform until buyer confirmation, enabling effective platform oversight and enforcement [24], [25].

A comparison of these platforms is summarized in Table I. Together, these four platforms cover major second-hand marketplaces across different regions, ranging from minimally mediated local classifieds to platforms with stronger transaction control. Their differences in platform intervention, target user populations, customer protection, and communication structure provide a useful basis for examining how scam behaviors vary across transaction environments.

B. Threat Model and Scope

We define an SHScam as a fraudulent activity in which a second-hand marketplace plays a substantive role in victim acquisition, trust establishment, interaction, or transaction execution or redirection. The adversary's objective is to obtain money, goods, account access, or sensitive information, regardless of whether a legitimate transaction is completed. For this purpose, adversaries employ a range of tactics to manipulate

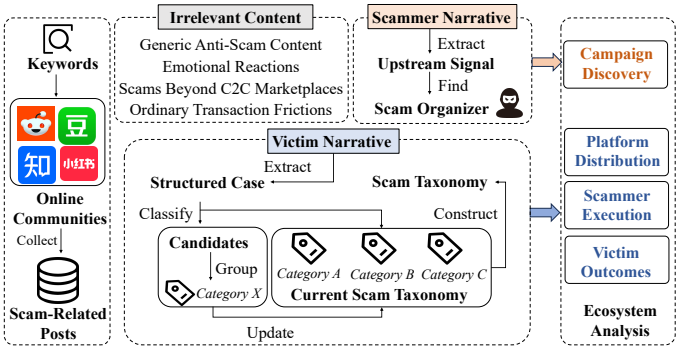


Fig. 2: Overview of LLM-assisted processing pipeline.

victims and evade platform oversight. In our threat model, adversaries can create or control marketplace-facing accounts, pose as buyers, sellers, or auxiliary roles, and interact with victims across both on-platform and off-platform channels. They exploit marketplace workflows—such as listing creation, negotiation, payment, shipping, refund, receipt confirmation—and may incorporate attacker-controlled infrastructure, such as phishing pages or fake payment interfaces, to carry out scams.

Our scope includes scams that originate from or critically depend on the marketplace at any stage of the attack. Scams that do not involve marketplace interaction fall outside our scope, even if they employ similar fraud techniques. This scope allows us to compare scams that occur on different platforms, while ensuring that all cases share a dependence on the second-hand marketplace ecosystem.

III. Methodology

To study the scam ecosystem on second-hand online marketplaces, we conducted a data-driven analysis using posts from online communities. While prior work [14], [15], [8], [9] has developed taxonomies for traditional scams, the diversity and rapid evolution of scams on second-hand marketplaces (SHScams) render existing taxonomies outdated and incomplete. Additionally, the descriptions of SHScams are often fragmented or ambiguous, making accurate characterization challenging. To address these issues, we leverage LLM-assisted structuring and a self-evolving framework to capture both established and emerging scam patterns. This section details the data sources, processing pipeline, and taxonomy construction procedures used in our study.

A. Data Collection

A comprehensive analysis of SHScams requires visibility into diverse real-world scam cases. However, finding them is non-trivial, as such scams are often disguised as legitimate listings on second-hand online marketplaces, while ground-truth data such as platform logs are generally not publicly available [26], [24]. To address this challenge, we follow prior work [14], [15] and take scam-related posts on public online communities as our entry point. These posts contain abundant victim reports while also revealing scam-related

signals beyond direct victim accounts, providing a broad and practical lens for studying the SHScam ecosystem.

Our study covers major second-hand platforms worldwide, including eBay, Craigslist, Facebook Marketplace, and Goofish [10], [11], [12], [13]. Given the geographical and linguistic divergence of their users, we collected reports from different online communities. Reddit [27] serves as the primary source for second-hand platforms used by Western users (i.e., eBay, Craigslist, and Facebook Marketplace), whereas Goofish-related reports were gathered from RedNote (Xiaohongshu) [28], Zhihu [29], and Douban [30], where Chinese users more commonly discuss marketplace scams. Across all sources, we collected only publicly accessible posts and recorded only the post URL, text, and timestamp. This process yielded 11,701 candidate posts for subsequent analysis. Details of data collection are provided below.

• *Reddit* is a global online forum organized into topic-specific subreddits. We collected scam-related posts from the group "r/scams" [31] and filtered for second-hand marketplace content using predefined keywords (e.g., "eBay," "Craigslist," "Facebook Marketplace," "FB Marketplace," and "FBM"). Posts published before April 2023 were collected from the Pushshift archive [32], while later posts were obtained from a community-maintained dataset hosted on Academic Torrents [33]. This yielded 9,086 candidate posts spanning October 2010 to June 2025.

• *RedNote, Zhihu, and Douban* are major Chinese online communities. Douban hosts dedicated anti-scam groups similar to Reddit's subreddits. We therefore collected data from four such groups [34], [35], [36], [37] using keyword filtering (e.g., "Goofish") to focus on second-hand marketplaces. On RedNote and Zhihu, scam-related content is not concentrated in dedicated discussion groups. We thus relied on keyword-based search, issuing multiple localized queries (e.g., "scams on Goofish," "Goofish scam," and "Goofish fraud") via MediaCrawler [38] to improve recall under their fuzzy search mechanisms [39]. All queries were issued in Chinese to align with the predominant user base. In total, we collected 2,615 Goofish-related posts from Chinese-speaking users, spanning from September 2016 to December 2025.

B. Data Processing

Posts collected via keyword-based queries are heterogeneous and often informal or ambiguous, spanning victim experiences, help-seeking questions, warnings, and other scam-related discussion without a concrete incident. We therefore applied semantic filtering to identify posts suitable for case-based analysis, followed by structured transformation to reduce inconsistencies caused by variations in expression.

Semantic Filtering. As shown in Figure 2, for each post, we applied an LLM-based classifier to analyze the content and infer contextual information. Based on this analysis, each post was assigned to one of three categories: (1) *Victim Narratives*, which describe concrete scam experiences from the victim's perspective, whether presented as retrospective narratives or help-seeking inquiries; (2) *Scammer Narratives*, which are written from the perspective of participants in scam operations, often illustrating how they were manipulated or involved in the fraud; (3) *Out-of-scope Content*, which does not describe a concrete second-hand marketplace scam incident, including generic anti-scam content, emotional reactions without specific incidents, scams unrelated to C2C marketplace contexts, and ordinary transaction frictions without clear malicious intent.

In the subsequent analysis, we focus mainly on *Victim Narratives* for the systematic measurement of SHScams, as they provide sufficient detail to reconstruct scam interactions. *Scammer Narratives* are retained as a secondary dataset for complementary upstream analysis (see §VI), while *Out-of-scope Content* is discarded. This step reduces noise while preserving both downstream incidents and upstream signals.

Structuring. For posts retained in the victim-narrative dataset, we leveraged an LLM to extract structured representations of scam cases, transforming unstructured narratives into structured data suitable for systematic analysis. Building upon prior work [40], we defined seventeen fine-grained attributes to characterize each case, including scammer role, traded item, payment channel, transaction stage, trigger event, scam mechanism, behavioral sequence, channel migration pattern, scam outcome, and victim loss (detailed in Appendix B). Stored as a JSON object, this representation captures the specific *scam pattern* employed by scammers in each case and serves as a standardized input for SHScam classification. Cases with four or more critical attributes marked as uncertain were considered *incomplete samples* and excluded from further analysis and taxonomy construction.

C. Scam Taxonomization and Classification

After structuring, we assigned each case to a scam category to support ecosystem-level measurement. This required a taxonomy that is fine-grained enough to capture marketplace-specific abuse, yet flexible enough to absorb previously uncatalogued variants. Existing taxonomies proposed by regulatory agencies (e.g., FTC [8] and FBI IC3 [9]) and prior academic studies [14], [15], [41] are intentionally broad and are not tailored to second-hand marketplace workflows. In addition, while they cover common fraud types such as fake checks [42] and advance-fee fraud [43], they are insufficient for capturing the diverse patterns of SHScams. We therefore adopted a self-evolving framework for scam classification and taxonomy construction [44], [45], [46]. As shown in Algorithm 1, the framework operates on structured case representations and expands the taxonomy only when recurring scam patterns cannot be explained by the current categories.

In-Context Classification. We classified each structured case using in-context learning [47]. Rather than operating on raw text, the classifier uses the extracted case attributes to reason about the underlying scam logic, especially scammer role, transaction stage, behavioral sequence and trigger events, which can reveal the structural patterns and causal relationships of the scams. This keeps classification grounded in mechanism rather than superficial transaction context. The taxonomy is initialized with an expert-defined

Algorithm 1: Scam taxonomization and identification.

Input: Seed taxonomy T_0; structured cases $\mathbf{A}$; confidence threshold $\tau = 0.7$
Output: Evolved taxonomy T; final labels $\mathbf{L}$

$T \leftarrow T_0$;
$\mathbf{P} \leftarrow \emptyset$; // All cases initially labeled as new category
$\mathbf{L} \leftarrow \emptyset$;
foreach *case* $a \in \mathbf{A}$ **do**
  $(c, v) \leftarrow$ LLM-CLASSIFY(a, T);
  $c \leftarrow \arg\max(v)$;
  $s \leftarrow \max(v)$;
  if $s < \tau$ **then**
    $(c', v') \leftarrow$ CROSSMODEL-VALIDATE(a, T);
    if $\max(v') > s$ **then**
      $(c, v) \leftarrow (c', v')$;
      $c \leftarrow \arg\max(v)$;
      $s \leftarrow \max(v)$;
  if $c \neq$ *new-category* **then**
    $\mathbf{L}[a] \leftarrow c$;
  else
    $\mathbf{L}[a] \leftarrow$ *new-category*;
    $\mathbf{P} \leftarrow \mathbf{P} \cup \{a\}$;
    if $s \geq \tau$ **then**
      $\mathbf{D} \leftarrow$ LLM-DISCOVER$(\{a' \in \mathbf{P} : s(a') \geq \tau\}, T)$;
      if $\mathbf{D} \neq \emptyset$ **then**
        $k \leftarrow$ LLM-SUMMARIZE$(\mathbf{D})$;
        $k \leftarrow$ HUMAN-VALIDATION(k);
        $T \leftarrow T \cup \{k\}$;
        foreach *case* $a' \in \mathbf{D}$ **do**
          $\mathbf{L}[a'] \leftarrow k$;
          $\mathbf{P} \leftarrow \mathbf{P} \setminus \{a'\}$;
foreach *remaining case* $a \in \mathbf{P}$ **do**
  $c \leftarrow$ MANUAL-REVIEW(a, T);
  if c *matches an existing category* **then**
    $\mathbf{L}[a] \leftarrow c$;
  else
    $k \leftarrow$ HUMAN-DEFINECATEGORY(a);
    $T \leftarrow T \cup \{k\}$;
    $\mathbf{L}[a] \leftarrow k$;
return $T, \mathbf{L}$

seed set T_0 (see Appendix A) derived from prior studies and official warnings [14], [15], [8], [9]. For each structured case $a \in \mathbf{A}$, the model produced a confidence vector $v = [s_{c_1}, s_{c_2}, ..., s_{c_n}, s_{\text{new}}]$, where s_{c_i} denotes the confidence for category c_i, and s_{new} denotes the confidence for a previously undefined new category. The case label is determined by the highest-confidence entry in v. To improve classification reliability, if this confidence score is below 0.7[1], we further re-evaluate the case via an independent model and retain the label with the higher confidence score across the two model outputs. Cases are then assigned to existing categories or routed to category discovery, enabling both classification into existing categories and the discovery of emerging scam patterns.

Self-Evolving Taxonomy Construction. The taxonomy construction follows a semi-automated iterative process designed to balance scalability with categorical rigor. Cases routed to *new-category* are placed in a candidate pool $\mathbf{P}$. We used a separate LLM to identify recurring patterns and derive candidate categories, considering only cases in the candidate pool with confidence scores above 0.7 to prevent low-confidence cases from influencing the taxonomy expansion. If a case shares a consistent pattern with an existing case in $\mathbf{P}$, the matching cases are grouped into a set $\mathbf{D}$, from which the LLM derives a new category along with its definition and pattern description. Following prior work on iterative taxonomy construction [48], [49], we included a human-in-the-loop step to verify that the new category does not overlap with existing ones. Once validated, the category is added to the SHScam taxonomy, and all associated cases are labeled and removed from $\mathbf{P}$. For the remaining long-tail cases in $\mathbf{P}$, primarily the low-confidence cases excluded from new-category generation, we conducted manual review to determine whether they should be merged with existing categories or formed new ones. All cases are ultimately assigned to categories, yielding the final labeling results $\mathbf{L}$. To ensure categorical clarity and consistency, we performed an expert-guided semantic aggregation as a post-processing step. This step consolidates the LLM-generated categories based on the scammer roles and underlying scam logic without altering the underlying category definitions. Our analysis finally yielded 43 SHScam categories, 35 of which emerged beyond the initial seed taxonomy. The resulting taxonomy is used throughout §IV-A, with full category definitions provided in Appendix A.

[1]We set the confidence threshold to 0.7 as a conservative criterion for flagging cases for cross-model review. The threshold served only this operational purpose and was not interpreted as a calibrated probability.

D. Implementation and Validation

Implementation. Our design is model-agnostic, and we used GPT-5 [50], a state-of-the-art large language model at the time of experimentation, as the primary model to support reliable analysis. For cross-model validation, we used Claude Opus 4.1 [51] to independently reassess the low-confidence cases. We employed two LLM-based components with distinct roles, one for data processing and classification and the other for category discovery. These components are integrated into a single pipeline that processes posts sequentially. To handle both English (77.65%) and Chinese (22.35%), we maintain language-specific prompt variants while preserving a consistent output format across languages. The prompts and full implementation are available at [52].

Validation. We conducted a post-hoc human validation to evaluate our data processing pipeline across four dimensions: filtering accuracy, category assignment, structured extraction, and taxonomy interpretability. To balance representativeness and analysis feasibility, we used a mixed sampling strategy to construct a validation dataset of 230 posts (2% of the full corpus). Specifically, we randomly sampled 130 posts from the 3,392 cases assigned to specific scam categories to ensure adequate coverage of categorized instances, and 100 posts from the full corpus of 11,701 posts to assess overall filtering performance. Two expert annotators, both with prior research experience in online scams, independently performed the assessments. Detailed annotator task descriptions and criteria are also publicly available at [52].

• *Filtering accuracy.* For each of the 230 posts, annotators assigned all applicable categories based on the classification standards defined in §III-B. Filtering accuracy was then evaluated specifically for *Victim Narratives* category, as correct assignment to this category is essential for taxonomy construction and downstream analyses. Inter-annotator agreement was

TABLE II: Validation results of the analysis pipeline.

Validation Target	Sample	Main Metric	Result
Case filtering	230 posts	Precision	100.0%
Category assignment	150 cases	Accuracy	94.0%
Structured extraction	150 cases	Rubric score (Mean)	4.46/5
Taxonomy design	127 cases	Category-fit rate	90.6%

high (Cohen's $\kappa = 0.91$), with disagreements resolved through discussion to establish the ground truth. The filter pipeline achieved 100.0% precision (95% Wilson CI: 97.5%–100.0%) and 93.1% recall (95% Wilson CI: 88.2%–96.1%), indicating a conservative but reliable filtering process.

• *Category assignment.* Among the 230 samples, 150 posts were assigned to categories, including 20 from the full-corpus sample. Annotators evaluated whether the model-assigned categories matched the expected scam mechanisms and were supported by the original narratives. Agreement between the experts was high (Cohen's $\kappa = 0.86$), and the assignment achieved 94% accuracy (95% Wilson CI: 89.0%–96.8%). The few mismatches mostly occurred in boundary cases whose underlying scam mechanism was present but not fully specified in the original narratives.

• *Structured extraction.* For these 150 categorized posts, we further evaluated the accuracy of the LLM's structured extraction. Annotators scored each output on a 5-point rubric[2], assessing whether it accurately captured key scam attributes. The mean score was 4.46/5 (median = 4), with 98% of samples scoring 4 or 5, indicating that the extracted representations faithfully reflected the original narratives.

• *Taxonomy interpretability.* To assess the semantic coherence and interpretability of the taxonomy, we manually selected up to three representative cases[3] for each category, yielding 127 cases[4] in total. The cases were selected to clearly reflect the key attributes and typical scam mechanisms of their assigned categories. Two annotators reviewed each category definition together with its representative cases and assessed whether the cases fit the category and were not better explained by another category. Overall, 90.6% of the reviewed cases were consistently well represented by their assigned categories, with only a few touching adjacent category boundaries, supporting the clarity of our category definitions.

In summary, these results indicate that our data processing pipeline enables the efficient analysis of scam reports and provides a solid foundation for further measurement analysis.

[2]Higher scores indicate that the structured representation is faithful to the original narrative, captures the key scam attributes, and preserves uncertainty without unsupported inference. Lower scores reflect missing core details, speculative additions, or distortion of the scam process.

[3]Representative cases were selected as those that (1) clearly reflected the key attributes and typical scam mechanisms of the category, (2) included a complete or near-complete sequence of relevant behaviors, and (3) were described in clear and unambiguous language to facilitate accurate annotation.

[4]Three samples were drawn per category (43 categories), except for categories with fewer than three instances, resulting in 127 cases in total.

IV. Landscape of SHScams

Our approach synthesizes fragmented observations into a unified SHScam framework, providing a systematic overview of scams on second-hand marketplaces. We finally identified 43 categories from 3,392 posts (1,143 from Chinese online communities and 2,249 from Reddit). Note that all the following analyses were conducted within our collected corpus. Given the partial visibility of victim reporting and the variation in reporting practices across online communities, the observed category richness and distributions reflect the diversity and coverage of scam reports in our discussion sources rather than the ground-truth diversity of scams in the wild.

A. Scam Category

As summarized in Table III, we organized the 43 categories into 10 groups according to the adversarial roles. Groups A–C (31.37% of cases) capture buyer-side abuse, Groups D–F (57.43% of cases) capture seller-side abuse, and Groups G–J (11.20% of cases) capture schemes that extend beyond direct buyer-seller interactions, involving shifted roles or objectives. Detailed category definitions are provided in Appendix A.

Finding 1: The structure of second-hand marketplaces gives scammers flexibility to impersonate either buyers or sellers and exploit any stage of a transaction, creating a highly diverse attack surface.

Buyer-Side Scams. Buyer-side scams (Groups A–C) account for 31.37% of cases, extending well beyond simple non-payment. Scammers exploit different stages of the transaction workflow to avoid payment, shift costs onto sellers, or abuse post-delivery protections. Group A (10.17%) targets the payment stage, taking advantage of weak platform mediation to rely on tactics such as forged payment proofs (A4) and reversible transfers (A3). Group B (14.50%) exploits the negotiable nature of second-hand trading to introduce artificial transaction barriers such as fabricated overpayments (B1) and fake payment failures (B5). Victims are then manipulated into making additional payments, either by refunding a supposed excess or by paying required fees such as the store-registration fee. Group C (6.69%) operates after delivery, using social engineering and platform policies to evade payment or recover funds. For example, scammers may claim product defects to obtain refunds while returning empty packages, leaving sellers without either goods or compensation (C1).

Seller-Side Scams. Seller-side scams (Groups D–F) constitute the majority of fraudulent activity, accounting for 57.43% of all cases. By controlling the listing and fulfillment process, sellers are better positioned to shape the transaction and thus have more opportunities to execute scams. Low entry requirements and weak verification for sellers further reduce the cost of malicious participation as sellers. Group D (36.56%) is the most prevalent pattern, where scammers take payment but fail to deliver, primarily through non-delivery (D1), fake shipment (D2), and misrepresentation (D3). Beyond delivery-stage fraud, Groups E (10.76%) and F (10.11%) target the payment stage with distinct strategies. Group E employs surcharge

TABLE III: The SHScam taxonomy organized by scammer roles (*: seed categories; : buyer-side scams; : seller-side scams; : transaction-deviating scams).

Group & Category	Mechanism	# Cases
A. Buyer-Side Payment Evasion		
A1. In-Person Nonpayment	meetup handoff → flee unpaid	42
A2. Counterfeit Cash Payment	counterfeit cash at handoff	8
A3*. Reversed Electronic Payment	reversible e-payment → reversed later	52
A4. Fabricated Payment Proof	fake receipt / fake platform notice	158
A5. Fake Check Payment	bad check → later rejected	85
B. Buyer-Induced Seller Payments		
B1. Fake Overpayment Refund	fake overpayment → refund "excess"	152
B2. Induced Gift Card Purchase	gift-card purchase → codes stolen	48
B3. Fake Pre-Sale Verification	fake inspection → verification fee	33
B4. Seller Shipping Deposit	shipping deposit requested from seller	16
B5. Order Unfreeze Fee	order frozen → pay unfreeze fee	243
C. Post-Delivery Buyer Abuse		
C1*. Post-Delivery Refund Abuse	refund abuse after receipt	162
C2*. False Non-Receipt Claim	false non-receipt after handoff	25
C3. Rental Non-Return	rented item kept	3
C4. Premature Code Release	code released before payment	37
D. Seller Non-Delivery / Misdelivery		
D1*. Non-Delivery After Payment	full payment → seller vanishes	412
D2*. Fake Shipment	fake tracking / empty parcel	227
D3*. Misrepresented Physical Goods	wrong, fake, or low-quality item	342
D4*. Invalid Digital Credentials	used / expired / invalid code	37
D5. Revoked Digital Access	access later revoked	44
D6. Misrepresented Virtual Goods	virtual item is bait or mismatch	28
D7. Untransferable Goods	ownership cannot be transferred	36
D8. Fake Rental Listing	fake home / fake availability	114
E. Seller-Induced Advance or Additional Payments		
E1. Pre-Transaction Service Fee	application / viewing fee first	19
E2. Fake Background Check Fee	sham credit → background check	68
E3*. Advance Shipping Fee	pay shipping before shipment	52
E4. Reservation Deposit	deposit to hold item	139
E5. False Payment Failure	payment failed → pay again	17
E6. Post-Payment Surcharge	extra fee after payment	41
E7. Post-Delivery Paywall	delivered item stays locked	10
E8. Task-Gated Delivery	tasks required before delivery	19
F. Instant Fund Interception		
F1. Early Receipt Confirmation	confirm receipt before delivery	76
F2. Gift Card Payment	pay by gift card / voucher	46
F3. Fake Transaction Platform	fake checkout / payment site	63
F4. Fake Escrow Claim	fake escrow / guarantee story	158
G. Relay / Intermediary Fraud		
G1. Relay Listing Fraud	copied listing → middleman relay	13
H. Marketplace-to-External Scam Pivots		
H1. Paid Training Job Offer	fake job with paid onboarding	20
H2. Task-Based Job Offer	marketplace lead into task scam	37
I. Third-Party / Proxy Exploitation		
I1. Search-Engine Fake Support	external fake customer service	1
I2. Proxy Recruitment	recruit victims as proxy sellers	19
I3. Unreimbursed Proxy Payment	errand / proxy payment request	10
J. Credential Theft and Phishing		
J1. Fake Interface Phishing	fake login interface	67
J2. Fake Redemption Page	fake code redemption link	11
J3. Verification Code Theft	social engineering for OTPs	202

extortion, demanding additional fees before delivery (E1–E4) or leveraging sunk costs to extract further payments after initial transactions (E5–E8). In contrast, Group F bypasses platform protections through attacker-controlled infrastructure, such as fraudulent payment interfaces (F3) or fake escrow services (F4), to ensure funds are transferred directly to the scammer.

Transaction-Deviating Scams. Beyond direct transactions, Groups G–J capture more complex patterns that move beyond buyer-seller interactions toward multi-party involvement or shifted objectives. Group G (0.38%) features scammers acting as both buyer and seller to intercept funds, while Group I (0.88%) involves third-party scammers unrelated to either role.

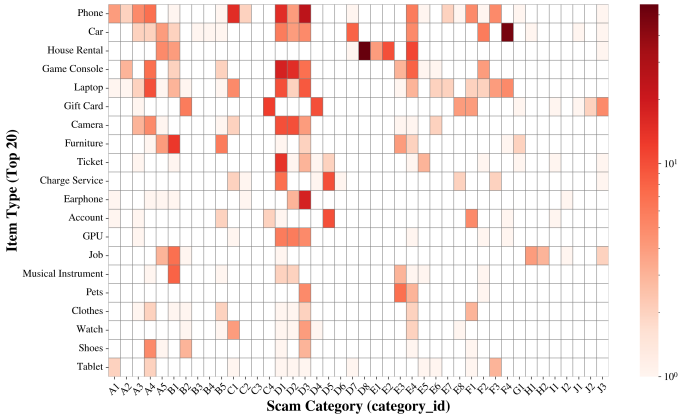


Fig. 3: Distribution of traded items across scam categories.

More critically, Group H (1.68%) uses the marketplace as a reputation proxy to initiate contact and redirect victims to external scams such as job scams. Group J (8.25%) shifts the target to digital credentials, with J3 (verification code theft) serving as a key entry point for account takeover and downstream exploitation. These patterns indicate that second-hand marketplaces are not only transaction venues but also infrastructure for initiating and facilitating downstream scams.

B. Common Victim-Facing Tactics

Despite the diversity of SHScam categories, many share common execution tactics, particularly those targeting victims to increase perceived legitimacy and reduce suspicion.

> **Finding 2:** Scams are often embedded in seemingly legitimate transactions, with scammers targeting victims through deliberate item selection and social engineering techniques to construct plausibility.

Scam Item Selection. Based on 1,170 posts mentioning traded items, Figure 3 shows that scam-related listings span both physical goods and virtual assets. However, this diversity is not random. Electronic products such as phones, computers, and game consoles appear especially often, suggesting that scammers prefer goods with broad demand and straightforward resale potential. The average listed price of phones, the most frequently traded item, is $419.51, which is consistent with typical second-hand market prices rather than an obviously suspicious bargain. High-value items such as luxury goods, jewelry, furniture, and musical instruments are also frequently involved in scams, aimed at maximizing profits. In addition, certain scam categories exhibit specific product preferences. For instance, E3 scams predominantly involve pets, while F4 scams are mainly associated with vehicles, as these items enhance the plausibility of scams.

Psychological Manipulation. Scammers exploit the initial trust offered by second-hand marketplaces and further manipulate victims via psychological tactics. Impersonation of institutional authority is the most common and direct strategy. In 13.41% of cases, scammers mimic official entities (e.g.,

TABLE IV: Success rates of commonly reported scams.

Category	Successful	Failed	Success Rate
D1. Non-Delivery After Payment	368	19	95.09%
D2. Fake Shipment	119	42	73.91%
D3. Misrepresented Physical Goods	267	34	88.70%
B5. Order Unfreeze Fee	55	186	22.82%
J3. Verification Code Theft	67	81	45.27%
F1. Early Receipt Confirmation	64	6	91.43%

customer support) by using forged emails or interfaces to legitimize their requests. In Category I1, scammers take a more proactive approach by advertising themselves as legitimate support services through search engines, targeting victims seeking help and broadening their reach beyond individual transactions. Moreover, scammers intensify manipulation through threat-based urgency, appearing in 13.97% of cases, with a high prevalence in Categories B5, D3, and C1. Additionally, 8.61% of cases, especially B5, B1, and E4, exploit repeated payment requests or escalating fees, increasing victims' sunk costs and reducing their likelihood of abandoning the transaction, thereby facilitating the scam. These patterns indicate that scammers combine authority cues with escalating pressure to systematically move victims from uncertainty to compliance.

C. Real-World Impact on Victims

Beyond the details of scam execution, victim reports from online communities also provide valuable insights into what happens after the scam. In this subsection, we analyze the consequences of scam incidents as reported by victims, including whether the scam succeeds, how much the victim loses, and whether any recovery is possible.

Finding 3: Scam success varies across categories and results in substantial real-world harm, with victims experiencing an average loss of $841 and a long-tailed distribution.

Success Rate. Not all reported cases in online communities result in successful scams, as victims may recognize malicious intent and terminate the transaction. After excluding cases without reported outcomes, the overall observed success rate is 59.47% (1,771/2,978 cases). Table IV shows the success rates of six commonly reported scam categories. *Non-Delivery After Payment* (D1), *Fake Shipment* (D2), and *Misrepresented Physical Goods* (D3) achieve high success rates, as victims lose control after payment and have limited ability to verify or reverse transactions. Similarly, F1 scams reach a high success rate (91.43%) by exploiting "confirm receipt" mechanisms, where victims are induced via phishing to prematurely release payment. By contrast, *Verification Code Theft* (J3) and *Order Unfreeze Fee* (B5) exhibit lower success rates, as they more clearly disrupt expected transaction logic prior to payment and are therefore easier to detect.

Scam Losses. According to victims' self-reports, 74.91% of cases involve direct monetary theft, often through goods seizure, payment fraud, or additional payment requests. The remaining scams focus on sensitive information or account access, which may not cause immediate financial loss but can lead to risks such as unauthorized charges or further scams. Since information loss is difficult to quantify, we focus on monetary losses. On average, victims reported a loss of $841.03, with a long-tailed loss distribution. Half of the cases reported losses under $238, while a small percentage resulted in significantly higher losses. The largest reported loss in our dataset involves a horse transaction worth $40,000. Figure 4 further presents the distribution of losses across scam categories. Scams in Groups B, F, H, and J exhibit markedly higher median losses, with Group B showing significant variability and a heavy upper tail. In contrast, Groups C and D manipulate the delivery or receipt of goods, resulting in one-time limited losses constrained by item prices.

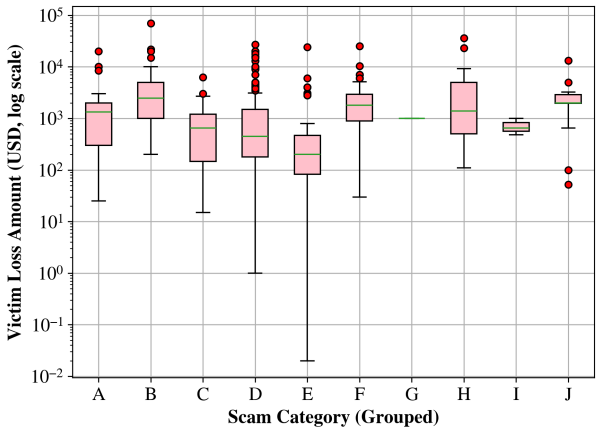


Fig. 4: Distribution of victim losses by category.

V. Cross-Platform Analysis

Based on our semantics-driven structuring process, the finalized dataset comprises 873 cases from Facebook Marketplace, 715 from Craigslist, 432 from eBay, and 1,143 from Goofish, with a small subset lacking explicit labels of second-hand platforms. In addition, 193 cases from other global marketplaces, including Gumtree [53], Flippa [54], OfferUp [55], and Kijiji [56], were retained through LLM-based semantic validation. These cases were initially identified via keyword matching, as users often described platforms by analogy (e.g., "similar to Facebook Marketplace"). These cases broaden the geographical scope and generalizability of our study and are therefore retained in the overall dataset and taxonomy analysis in §IV. However, we exclude them from the cross-platform comparisons in this section because their small sample sizes are not sufficiently representative.

A. Platform Distribution

Finding 4: Reported scams vary across marketplaces, highlighting how platform features (e.g., transaction models, user habits, and policy evolution) shape which scam categories emerge and gain prominence over time.

TABLE V: Top-5 scam categories on different marketplaces.

Marketplace	Top 5 Categories	Rate
Facebook Marketplace	D1. Non-Delivery After Payment	18.79%
	J3. Verification Code Theft	11.34%
	A4. Fabricated Payment Proof	8.93%
	B5. Order Unfreeze Fee	8.59%
	D3. Misrepresented Physical Goods	8.13%
eBay	D2. Fake Shipment	29.63%
	D3. Misrepresented Physical Goods	14.12%
	D1. Non-Delivery After Payment	12.50%
	C1. Post-Delivery Refund Abuse	10.42%
	A4. Fabricated Payment Proof	8.33%
Craigslist	B1. Fake Overpayment Refund	15.66%
	D8. Fake Rental Listing	12.17%
	F4. Fake Escrow Claim	12.17%
	J3. Verification Code Theft	9.09%
	A5. Fake Check Payment	8.95%
Goofish	D3. Misrepresented Physical Goods	14.26%
	B5. Order Unfreeze Fee	13.56%
	C1. Post-Delivery Refund Abuse	9.97%
	D1. Non-Delivery After Payment	8.66%
	F1. Early Receipt Confirmation	6.30%

As shown in Table V, reported scam categories differ substantially across marketplaces despite their shared C2C nature, reflecting scammers' adaptation to marketplace positioning. Facebook Marketplace is dominated by high-volume, general-purpose scams such as *Non-Delivery After Payment* (D1), as social network integration fosters perceived trust that lowers barriers for straightforward fraud. *Fake Rental Listing* (D8), *Fake Overpayment Refund* (B1), and *Fake Check Payment* (A5) are particularly prevalent on Craigslist, consistent with its focus on local, loosely mediated exchanges. As the primary second-hand marketplace in China, Goofish serves as a multi-purpose marketplace and exhibits the most diverse scam landscape in our dataset (38/43 categories), whereas eBay covers the fewest (29/43) and shows a much narrower scam profile. This narrower distribution is likely attributable to eBay's relatively standardized product transaction model, which limits non-standardized listings (e.g., job or rental ads) that serve as entry points for scams in Group H.

Marketplace differences also reflect user habits. In particular, as electronic payments dominate everyday transactions in China, scam categories that rely on cash or checks, such as *Counterfeit Cash Payment* (A2) and *Fake Check Payment* (A5), are largely absent on Goofish. Scam distributions also evolve over time, shaped by platform growth and policy changes. While timestamps record report time rather than exact incident time and not all scams are reported, they still provide a useful signal of temporal trends. As shown in Appendix Figure 10, the launch of Facebook Marketplace in 2016 created new opportunities for scams, with reported incidents exhibiting an upward trend, reflected in increasing report density and the emergence of new scam categories (e.g., A4, D1, H2). Similarly, the expansion of eBay's Money Back Guarantee in 2018 [57] enabled broader exploitation, leading to a surge in reported *Post-Delivery Refund Abuse* (C1) cases.

B. Platform-Shaped Scam Tactics

Second-hand marketplaces support SHScams while also deploying various safeguards to constrain them, such as escrow-based payment. In response, scammers employ tactics to bypass these restrictions and preserve opportunities for fraud, in parallel with victim-facing tactics targeting users (see §IV-B).

Finding 5: Scammers move interactions off-platform and strategically evolve new variants in response to platform safeguards to avoid oversight and restrictions imposed by second-hand marketplaces.

Escaping Platform Oversight. Second-hand marketplaces support flexible negotiation, and scammers exploit this feature to move interactions outside platform oversight. In our dataset, 2,231 cases (65.77%) involve off-platform behavior. Even when in-platform messaging is available, scammers frequently redirect victims to WhatsApp [58], WeChat [59], QQ [60], email, or SMS, thereby moving conversations beyond marketplace visibility and dispute channels. The reverse pattern also appears: in some scams, victims are first identified on social media or other services and then redirected to second-hand marketplaces to borrow the legitimacy of a familiar platform. For example, certain B5 scams move sellers from social media to Goofish, while some F4 scams redirect victims to eBay. Thus, platform migration is not only a way to evade oversight but also a strategy to select the venue that best supports the next stage of the fraud. It is often accompanied by a shift to less traceable and harder-to-reverse payment channels, with scammers directing victims to peer-to-peer services such as Zelle, Venmo, Cash App, and Apple Cash.

Adapting to Platform Protections. Scammers do not simply abandon fraud when platforms introduce stronger safeguards. Instead, they adapt by shifting toward variants that remain compatible with platform workflows while still achieving similar fraudulent outcomes. As illustrated in Table V, within our reporting corpus, straightforward *Non-Delivery After Payment* (D1) is less frequently reported on platforms such as eBay and Goofish, where transactions are more structured and safeguarded. In their place, attackers more often adopt superficially compliant tactics such as *Fake Shipment* (D2) and *Misrepresented Physical Goods* (D3), which preserve the appearance of legitimate transactions while evading enforcement. Likewise, scammers exploit protection mechanisms directly to develop new scam patterns. This is reflected in scams such as *Post-Delivery Refund Abuse* (C1) and *Early Receipt Confirmation* (F1), which leverage refund, receipt, or dispute workflows to capture funds or retain goods. These categories suggest tactic substitution, with safeguards suppressing simpler scams while enabling more adaptive variants.

C. Platform Effects on Scam Recovery

Finding 6: Platform mediation plays a critical role in loss recovery. Although recovery is often limited, platforms with built-in protections offer higher recovery rates.

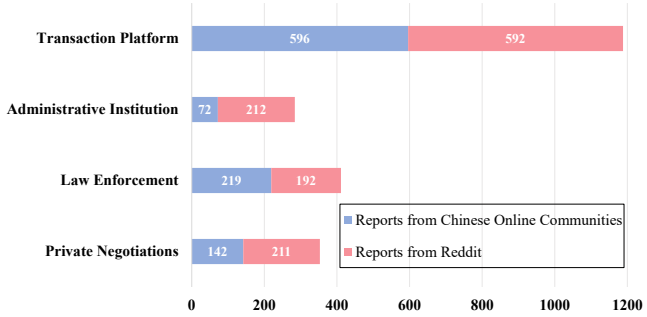


Fig. 5: Post-scam recovery measures taken by victims.

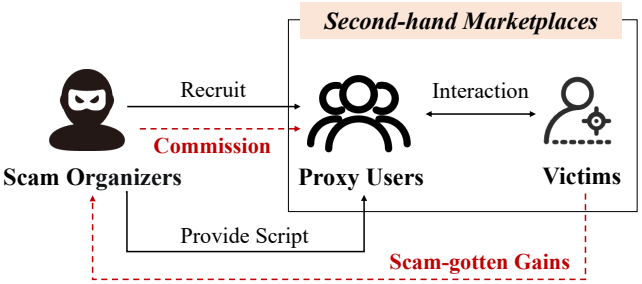


Fig. 6: Structure of the proxy-driven scam model.

Among all self-reported cases, 52.06% (1,766 cases) described attempts to recover losses, with a higher reported proportion on Goofish (63.69%, 728/1,143). As shown in Figure 5, victims primarily rely on marketplaces and payment platforms (e.g., PayPal and banks) to reverse transactions, often supplemented by formal reports to authorities (e.g., FTC [8]). In some cases, private negotiations are useful to resolve disputes, particularly on Goofish, which facilitates this process by providing a counterparty-disclosure mechanism[5]. Despite these efforts, recovery outcomes remain limited. Among the cases that attempted recovery, only 27.45% fully recovered their losses, while 4.59% achieved partial recovery. Recovery success varies by scam category, with relatively high recovery rates in Categories C1, D2, E4, and F1. Notably, recovery outcomes are generally better when scams remain within platform-mediated transaction workflows, where support mechanisms such as on-platform payment, transaction records, and dispute-resolution channels may still be available. Specifically, off-platform scams show lower recovery rates (27.76%) than on-platform scams (34.28%), suggesting that recovery becomes more difficult once victims move beyond platform-controlled processes. Moreover, platforms with stronger built-in transaction controls, like Goofish (39.05%) and eBay (36.48%), show higher recovery rates than less mediated marketplaces such as Craigslist (19.82%) and Facebook Marketplace (22.27%).

VI. Proxy-Driven Scam Campaigns behind SHScams

As discussed above, §IV and §V characterize SHScams based on victim-facing reports. In this section, we focus on 88 scammer-authored posts identified during data filtering. Although these posts were excluded from our primary victim-centric taxonomy, they provide valuable insights that are largely absent from victim-reported cases. In particular, these posts always describe participants' roles and experiences of being exploited as scammers. Such evidence suggests that some individuals are recruited into fraudulent operations rather than acting independently, pointing to the presence of upstream organized groups coordinating these activities.

[5]When disputes arise, users can submit supporting evidence to request the disclosure of the counterparty's verified contact information.

Finding 7: A subset of SHScams follows a proxy-driven campaign model that separates recruitment, marketplace deployment, and monetization.

Based on the content of these posts, we refer to this pattern as the *proxy-driven scam model*, where organizers recruit intermediaries to act on second-hand marketplaces, while retaining control over communication scripts and payment-harvesting infrastructure. As shown in Figure 6, proxies act as visible counterparties by posting listings or initiating contact, often without awareness of the underlying fraud and earning small profits by completing assigned activities. Once exposed, proxies absorb much of the risk and recognize their exploitation, leading them to post complaints in online communities. By using proxy accounts as visible front-ends, upstream organizers shift detection risks while scaling scam operations through accounts with established histories. As platform defenses tighten and account creation becomes more costly, particularly on platforms with real-name registration such as Goofish, this strategy becomes increasingly attractive. Among the 88 scammer-authored complaint posts, 29 were observed on Goofish, while the remainder appeared on Western-centric marketplaces, indicating that this proxy-driven model generalizes across platforms. These posts provide the signals necessary for our upstream analysis rather than for estimating the prevalence of this scam model.

Compared with *Proxy Recruitment* scams (I2), which also involve proxies acting as sellers, the intermediaries in I2 are the primary victims. In contrast, in this model, proxies directly interact with and target downstream users, effectively assuming the role of the scammers.

A. Empirical Validation of Proxy-Driven Scam Campaigns

Based on scammer-authored complaint posts, scam organizers recruit intermediaries directly through popular messaging apps such as Telegram, WhatsApp, and QQ. However, many recruitment activities occur in private or semi-closed channels, so our observations are limited to publicly accessible sources, such as Telegram where recruitment posts are openly shared. Specifically, we searched Telegram for recruitment posts using proxy-task keywords (e.g., "part-time posting," "account rental," etc.) via a third-party Telegram search engine [61]. This process identified 24 channels that contain different

"Goofish listing— clean and legitimate posting work. You'll be posting items like phones, tablets, and other electronics. Zero risk, earn 90RMB per day. Just post 50 listings, and you're done for the day. An advance of 20RMB is available after completing the listings. Contact us to get started."

Fig. 7: Recruitment advertisement published on Telegram.

recruitment posts targeting second-hand marketplaces, collectively reaching a total of 97,202 subscribers. The identified posts typically framed the work as low-risk, commission-based tasks and recruited proxies to pose as sellers or buyers, involving them in scam operations unintentionally. We next present a detailed analysis by proxy type.

Finding 8: By supplying specific content and scripts, upstream organizers enable downstream actors to facilitate the scalable execution of scams, ultimately producing real-world victim-facing cases.

Seller-Side Proxy Campaigns. Among the observed recruitment channels, 17 involved the recruitment of seller-side proxies, as illustrated in Figure 7. In these posts, the proxies impersonate sellers and post specified listings as required. To further examine these recruitment practices, we conducted exploratory interactions with a purposive sample of recruiters, selected to capture variation in assigned posting tasks, including goods types and instructions provided to proxies. These goods primarily included consumer electronics, as well as other high-demand goods such as concert tickets and membership-based products. To prevent harm, we did not disclose the organizers' Telegram account details or publish any listings (see the Ethical Considerations section for details).

• *Consumer electronics.* Across the observed channels, 13 focused on electronic products. To screen potential proxies, organizers requested screenshots of their Goofish accounts to verify that they had sufficient reputation. Proxies were then instructed to install WeBuy Album[6], which provides reusable product images and descriptions. Each listing had to include both the product descriptions and organizer-controlled contact information (e.g., a QQ account identifier), as shown in Appendix Figure 11. Proxies were explicitly told not to engage with buyers on the platform, ensuring that all subsequent interactions occurred off-platform.

To observe downstream behavior, we contacted the next-level scammer using the embedded contact information as a prospective buyer. After brief interaction, the scammer shared a purchase link that mimicked Goofish but hosted on a non-official domain. Although we did not complete the transaction for ethical reasons, inspection of the page revealed a payment-failure interface (Figure 8), indicating that payments would be captured by the scammers while falsely presented as failed. The scammer would not deliver any product and

[6]WeBuy Album is publicly available on official app stores such as Google Play [62] and the App Store [63]

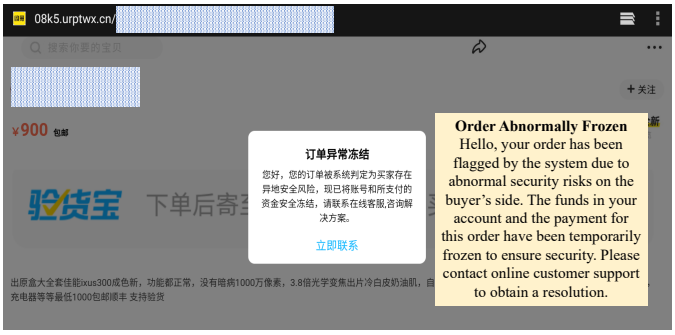


Fig. 8: Fake order notification sent to the victim.

could potentially solicit additional payments. This chain is consistent with *Fake Transaction Platform* (F3) and matches 19 victim-reported F3 cases in our dataset, in which victims are redirected to QQ-based communication and subsequently deceived into making payments on fraudulent pages.

• *Concert tickets.* A similar workflow was observed in one channel focused on concert tickets, with minor variations in implementation. Instead of using WeBuy Album, organizers directly provided proxies with ready-to-use images and descriptions, typically aligned with currently popular events. The listing content itself appeared legitimate and did not exhibit obvious anomalies. Proxies were also given communication scripts and instructed to redirect buyers off-platform by sharing a QQ contact, framed as a "ticketing agent" for order processing and ticket issuance. Victims were thus guided to QQ-based channels, where they were presented with links to fraudulent ticketing sites. The resulting workflow also corresponds to the *Fake Transaction Platform* (F3) pattern, and is consistent with four cases observed in our dataset.

• *Membership services.* We also observed a parallel seller-side variant in three channels that instructed proxies to advertise discounted vouchers or subscription services (e.g., Sam's Club day passes and QQ Music subscriptions). Organizers provided pre-prepared listing content, typically involving digital vouchers, along with scripted responses to guide buyer interactions. Proxies were instructed to restrict purchases to Apple users, after which they would share a designated fraudulent "redemption" link. In practice, this link led to a phishing page designed to steal users' Apple ID credentials, which were subsequently used to make unauthorized charges. This pattern is consistent with *Fake Redemption Page* (J2) and aligns with 11 J2 reports.

Buyer-Side Proxy Campaigns. Similar to seller-side schemes, buyer-side proxies are recruited to impersonate buyers and provided with communication scripts for approaching sellers on second-hand marketplaces. Their primary purpose is to funnel victims to upstream organizers rather than complete transactions. Among the Telegram search results, we identified seven Telegram channels that recruit such proxies.

Given the absence of entry requirements as buyers on second-hand marketplaces, the organizers publicly publish detailed, step-by-step guides within their channels to attract as many proxies as possible. These guides instruct proxies to

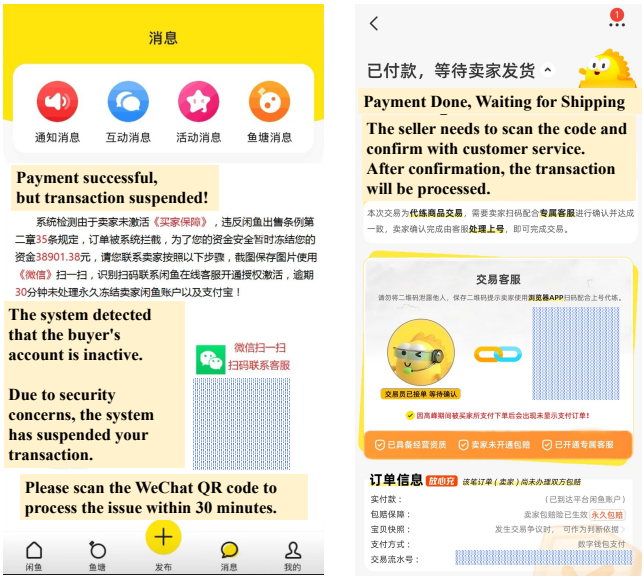


Fig. 9: Examples of order-freeze notification.

independently locate and engage with target sellers, typically focusing on newly registered accounts with no transaction history, who are more likely to be unfamiliar with standard transaction procedures. During interactions, proxies follow scripted instructions to simulate a payment attempt and subsequently deliver standardized "order-freeze" notifications. As shown in Figure 9, these messages include QR codes that direct victims to fraudulent support or payment pages, where they are prompted to pay "verification fees" or "store registration fees" to unfreeze a non-existent payment. This workflow represents a QR-code-based subtype of *Order Unfreeze Fee* (B5) observed in 110 reported cases, demonstrating the real-world prevalence of this scam pattern.

B. From Organized Campaigns to Detectable Signals

The alignment between observed recruitment workflows and victim-reported categories is mechanism-level and does not attribute any specific case to a particular recruiter or campaign. Beyond revealing organizational structure, the proxy-driven scam model generates reusable artifacts that propagate into marketplace listings and can provide an artifact-level link between upstream recruitment materials and potential downstream listings. Unlike reactive approaches based on user complaints, these artifacts allow for proactive detection, particularly in seller-side scams involving electronic products, where listing content exhibits distinctive off-platform redirection patterns, rather than relying on private interactions.

Finding 9: Upstream recruitment artifacts can be linked to active downstream scam deployment, exposing a broader campaign surface than victim reports alone.

Leveraging this insight, we used 90 images associated with 13 upstream-provided products from WeBuy Album as seed queries in Goofish's image-based search. Unlike text-based queries, which suffer from semantic ambiguity and broad results, product images serve as more precise and direct identifiers in the marketplace. This process identified 235 related listings from 32 accounts, all reusing identical materials and carrying explicit off-platform redirection signals in either images or descriptions. By examining posting activity, we found that these accounts posted multiple products of the same category within a single day, consistent with distributed task instructions rather than normal user behavior. Crucially, the listings identified via upstream-provided artifacts far outnumber the corresponding victim reports, indicating that downstream accounts provide only a partial view of ongoing scam activity and highlighting upstream signals as an effective means to reveal latent scam campaigns.

Additionally, we observed a rapid turnover of products and accounts within three-day periods. During this window, two shops were penalized, three products were marked as sold, and follow-up queries revealed four new shops that posted an additional 49 products. This pattern indicates systematic and recurring campaign deployment, resulting in sustained user exposure to scam risks.

VII. Discussion

Our work reveals a scam ecosystem in which platform design and transaction workflows shape execution, while upstream organizers supply the content and scripts that drive downstream scams. In this section, we discuss how these observations translate into practical implications for mitigation and summarize the limitations of our study.

A. Mitigation

Our findings highlight three key directions for mitigating scams on second-hand marketplaces.

Recognizing Workflow Deviations as Warning Signals. Based on our analysis, the most actionable warning signs in SHScams are often procedural rather than price-based. Many scams are embedded in otherwise plausible transactions involving ordinary goods, realistic price points, and seemingly legitimate interactions. What distinguishes them is the manipulation of key workflow steps, such as unexpected payment failures, deliberately fabricated overpayments, premature receipt confirmation, or requests for additional verification or service fees. For example, *Order Unfreeze Fee* (B5) induces additional payments, while *Early Receipt Confirmation* (F1) and *Post-Delivery Refund Abuse* (C1) exploit escrow release and refund workflows, respectively. This suggests that users should watch for workflow deviations rather than relying only on static content cues. In practice, requests for extra payments, specific payment channels, premature receipt confirmation, or off-platform customer communications should be treated as warning signs to terminate the transaction.

Evolving Platform Protections against Scammer Adaptation. Platform-provided user protections such as escrow-based payment mediation and refund workflows are critical for reducing fraud and improving loss recovery. However, as

scammers adapt to exploit remaining workflow gaps, platforms must continuously refine protections. For example, shipment evidence should not be treated as trustworthy based solely on tracking numbers. Refund systems should incorporate stronger item-return verification to prevent empty-package or item-switching abuse, and receipt-confirmation flows should include clearer warnings and stronger release controls. More broadly, platforms should assess protections not only by whether they block known scams, but also by whether they create new abuse surfaces elsewhere in the transaction lifecycle.

Monitoring Upstream Signals for Early Scam Detection. Our campaign analysis suggests that some second-hand marketplace scams are better understood as organized operations rather than isolated incidents. In these cases, the suspicious marketplace account is often only a disposable proxy, while the upstream organizers, reusable materials, and coordination channels remain intact. As a result, enforcement based solely on downstream victim reports, such as removing listings or suspending accounts based on post-hoc user reports, is often insufficient. This points to the need for campaign-oriented monitoring rather than purely case-by-case moderation. Platforms can incorporate upstream signals from external channels, such as Telegram, to identify scam activity before it causes real harm to users. Repeated listing images, fixed external contact details, rapid listing of similar products, and other proxy-style patterns can all serve as indicators of organized deployment, enabling earlier detection of scams on second-hand marketplaces.

B. Limitations

Our work provides the first systematic analysis of the SHScams ecosystem through scam reports from online communities. Nevertheless, several limitations should be noted.

Report-Based Data Source. Our dataset is derived from user-generated scam reports and is therefore subject to incomplete reporting, fabricated claims, subjective exaggeration, and variation in how incidents are described across communities. To improve data quality and reliability, we applied a series of semantic filtering and validation steps to remove low-quality reports and mitigate noise. We also interpret frequency-based results as observations of the reporting corpus rather than population estimates to avoid overinterpreting the data. Accordingly, our findings characterize broad trends, recurring patterns, and organizational structures in the observable SHScams ecosystem rather than providing precise reconstructions of individual incidents.

Data Coverage and Collection Scope. Our data collection focuses on four major second-hand marketplaces, supplemented by additional platforms identified through LLM-based semantic analysis, providing broad regional coverage of the observable SHScams ecosystem. In addition, our data were collected from a limited number of public online communities, and the specific search mechanisms constrain both the volume and the modality of retrievable data, limiting our analysis primarily to textual content. These coverage limitations may affect the observed frequencies of scam categories and their platform-level distributions. Nevertheless, our dataset represents a best-effort sample rather than a complete census of scam activities, and still yields meaningful empirical insights into the SHScams ecosystem.

LLM-Assisted Analysis. To facilitate scalable analysis of large volumes of unstructured scam reports, we utilized LLMs at various stages of our pipeline. While these models excel in semantic understanding, their outputs may be affected by hallucinations or contextual misinterpretation, potentially introducing inaccuracies. To mitigate these risks, we incorporated several safeguards into the analysis pipeline. Low-confidence cases were handled conservatively through cross-model validation, and only high-confidence cases were allowed to influence the evolution of the scam taxonomy. Also, manual inspection was applied to support the stability and reliability of the processing pipeline. As model capabilities continue to improve, this limitation is expected to be further alleviated.

Upstream Recruitment Channels. Our upstream analysis focuses on publicly accessible Telegram channels, which primarily targeted Goofish. Although proxy-related complaints also appeared on Western-centric marketplaces, we did not identify corresponding public recruitment channels. Additionally, scammer-authored posts also suggest other upstream recruitment channels beyond Telegram, but these activities often rely on direct messaging and lack publicly accessible tutorials, making large-scale collection difficult. Despite this limited coverage, our upstream analysis supports the existence of the proxy-driven scam model. Our goal is to document this phenomenon, not to estimate its prevalence. Future work could use automated conversational agents to expand coverage and enable more systematic measurement across channels.

VIII. Related Work

Scams in Online Ecosystems and Digital Commerce. Scam activity has become increasingly diverse, with scam entry points expanding beyond traditional channels such as phone calls [64], [65] and SMS [66], [67], [68], [69] to more online vectors, including private messages [70], public comments [71], and search engines [72]. Existing research generally adopts a scam-centric perspective, focusing on specific scam families such as technical support scams [73], [74], [75], online survey scams [76], and giveaway scams [77], [78], with the aim of characterizing their operational patterns and developing the corresponding defenses. Scams in digital commerce ecosystems have also been extensively studied, with scammers often initiating contact through seemingly legitimate transactions. Early research by Glickman introduced the classic Nigerian scams [6]. Later studies have systematically examined a wide range of scams, including Craigslist buyer scams [5], [79], refund fraud [80], pet scams [7], and fraudulent e-commerce websites [81], [82], [83], revealing the diversity of fraudulent strategies. However, prior work typically centers on a single scam type or a single platform. In contrast, our work offers a cross-platform perspective on scams within second-hand marketplaces. These C2C environments, characterized by weaker mediation, more flexible roles, and

diverse transaction workflows, introduce more opportunities for scams compared to traditional e-commerce.

User-Generated Data for Security Analysis. User-generated content on online communities has become an important data source. Prior work has leveraged forum data to study user perceptions and experiences, treating the content posted by users as an expression of their thoughts. For example, Li et al. [84] analyze Reddit discussions to understand attitudes toward security and privacy risks in smart homes. Similarly, Wei et al. [85] study help-seeking and help-giving on online communities in the context of image-based sexual abuse, while Tseng et al. [86] analyze online infidelity forums to understand the tools and tactics used in intimate partner surveillance. More recently, Oak et al. [14] and Bouma-Sims et al. [15] analyze scam-related posts on Reddit, highlighting how these online communities capture victim experiences, peer advice, community judgment, and collective sensemaking. Beyond capturing perceptions and discourse, user-generated content on online communities can also serve as a form of proactive threat intelligence, as demonstrated by prior work [87] leveraging Twitter data to identify and analyze SMS spam campaigns. Building on this, we focus on scam-related discussions on online communities and move from discourse analysis to ecosystem-level measurement. Instead of analyzing how users discuss scams [15], [14], we extract both victim-reported incidents and scammer-authored signals to characterize the scam ecosystem in second-hand marketplaces across platforms.

IX. Conclusion

Our work presents the first systematic measurement of scams on second-hand marketplaces from an ecosystem-level perspective. By automatically analyzing 3,392 scam cases extracted from public posts on online communities, we constructed a fine-grained taxonomy comprising 43 scam categories across four major platforms. Leveraging the open transaction nature of second-hand marketplaces, SHScams exhibit diverse patterns. Scammers can flexibly impersonate sellers or buyers, exploit multiple stages of the transaction process to execute scams, and even engage in non-transaction scams. We further examine cross-platform patterns and scammers' adaptations to platform constraints. Beyond victim reports, we identify a proxy-driven scam model connecting upstream recruitment, marketplace deployment, and downstream victimization, demonstrating the potential of upstream evidence for early detection. In general, our study contributes to a broader understanding of the scam ecosystem of second-hand platforms and provides insights for mitigating such cybercrime.

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Ethical Considerations

This section outlines the ethical considerations of our study. We first present a stakeholder analysis, together with the impacts of our study on them. We then discuss the mitigations implemented to minimize ethical risks and conclude with a justification for conducting this research.

Stakeholder Analysis. Our research involves five primary stakeholder groups, consisting of scam reporters (including victims), SHScam cybercriminals and associated proxies, online second-hand platforms, online communities, and researchers. Here, we articulate the role of each stakeholder and the potential impacts of our research on them.

- *Scam reporters* voluntarily share their experiences, which serve as a critical data source for our analysis of SHScams. While public, these reports may contain sensitive personal identifiers (e.g., names, contact details, or financial information) related to victims or scammers. To avoid the exposure of such sensitive information, we did not collect account information of reporters and de-identified all collected content before data processing. We strictly ensured that no raw data was further disseminated to prevent any secondary exposure or negative impact on the reporters.
- *SHScam cybercriminals and associated proxies* may be referenced in victim reports on online communities. To prevent potential harm from exposing personally identifiable information, we do not attempt to identify or target any specific individuals. In close collaboration with our institution's institutional review board (IRB), we crafted comprehensive guidelines to govern our interactions with scam organizers (e.g., requesting only task information from organizers without attempting to deanonymize their identities). These guidelines were designed to construct a robust legal and ethical framework, thereby minimizing any potential risk of harm.
- *Online second-hand platforms* are struggling to combat scams. Our study proposes a new methodology to automate scam analysis and monitor the evolution of SHScams. Our findings and public artifacts provide deep insight into this type of cybercrime, supporting platforms in improving scam detection and prevention mechanisms.
- *Online communities* provide a medium for information exchange. We collected data exclusively from publicly accessible posts using public APIs or publicly available tools. We did not collect private content or user account information. As a result, our data collection practices do not infringe upon the rights or operations of the communities involved.
- *Researchers* were exposed to a large volume of distressing scam narratives during the study. To mitigate vicarious trauma [88] and stress, we conducted regular peer-support conversations to support the mental well-being of researchers. Additionally, we created an anonymous Telegram account to maintain safety during interactions with the upstream scam organizer and generated a fabricated Goofish account screenshot with a high reputation score to gain the organizer's trust.

Mitigation. Our study was reviewed and approved by our institution's institutional review board (IRB). The study was

designed to construct a robust legal and ethical framework for data collection, researcher interaction, data processing, and reporting to minimize potential risks to all involved parties.

• *Methodological Mitigation*. For data collection, we accessed only publicly accessible content and enforced strict rate limits across all online communities. We retained posts and their publication timestamps but excluded user account information. All collected data were de-identified before analysis and stored on secure devices. To investigate upstream recruitment, we contacted selected Telegram users only when necessary to clarify their recruitment workflows. We kept these interactions brief and ended each interaction once the workflow was understood. We did not request or transfer money, disclose or retain personally identifiable information, accept tasks, publish listings, or participate in scam activities. Importantly, our study focuses on measuring existing scam activity and does not propose or demonstrate any new scams.

• *Responsible Disclosure*. We have responsibly disclosed our findings to affected online second-hand markets, including eBay, Craigslist, Facebook Marketplace, and Goofish. For Goofish, we additionally reported potentially fraudulent listings identified by matching marketplace images with materials from upstream recruitment channels. We also informed WeBuy Album about the misuse of their platform as a repository for scams. So far, Goofish actively responded to our reports and requested additional information, while the other platforms and WeBuy Album had not responded.

Justification for Research. To evaluate the ethical considerations and potential risks associated with our study, we conducted a critical analysis informed by the principles outlined in the Menlo Report [89] and the Cybersecurity Research Ethical Frameworks [90]. Consistent with previous cybercrime studies [91], [92], we believe the societal benefits of our work outweigh the minimal risks involved. Specifically, this study presents a large-scale measurement study of scams on second-hand marketplaces. We reveal a diverse ecosystem shaped by second-hand marketplaces and identify a proxy-driven scam model that organizes scam operations. Along with the extensive mitigation measures implemented in our study, we can conclude that conducting this research is ethically justified and socially beneficial.

ARTIFACT APPENDIX

A. Description & Requirements

To support reproducibility and transparency, we made our dataset publicly available on Zenodo. This includes 3,392 processed cases, each annotated with a unique identifier, structured LLM output, assigned scam category, and confidence score. To mitigate the potential risks of re-identification or harassment to individuals associated with posts, we excluded original post URLs and associated personal information from the released data. Additionally, we released all prompts, code, and annotation criteria used in the analysis and validation processes to facilitate reproduction and extension of our findings.

1) How to access: The artifact is distributed through the Zenodo record https://doi.org/10.5281/zenodo.22795421. The archive contains the following directories and files:

- `ProcessedData/`: the processed scam cases derived from online communities;
- `LLMPrompt/`: the implementation of an LLM-assisted automated analysis pipeline;
- `AnalysisCode/`: analysis scripts used for quantitative measurement and the validation criteria;
- `README.md`, `LICENSE`, and `requirements.txt`.

2) Hardware dependencies: No specialized hardware is required. The artifact can be inspected and executed on a general-purpose computer. No GPU is required.

3) Software dependencies: The artifact requires Python 3.9 or later and the packages listed in `requirements.txt`, including `pandas` and `matplotlib` for quantitative analyses and `openai` for LLM access.

B. Artifact Installation & Configuration

After extracting the archive, initialize the environment from the artifact root as follows:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

The analysis scripts can then be run directly from the artifact root without additional configuration.

Executing the LLM pipeline in `LLMPrompt/` requires configuring the model endpoint and API credentials in the provided implementation and supplying at least one source post as input. We do not include the original source posts in the artifact because releasing them may raise privacy, copyright, and research-ethics concerns. Instead, source posts can be independently collected from public online communities following the data-collection procedures described in the paper.

C. Experiment Workflow

The released components can be used independently. For the complete workflow, users can first run the LLM-based data-processing pipeline in `LLMPrompt/` using independently collected source posts:

1) collect source scam posts following the procedures described in the paper;
2) configure the model endpoint and API credentials, and run the corresponding script in `LLMPrompt/`;
3) use the resulting structured cases for downstream analysis.

To reproduce the main downstream analysis, users can directly use the released processed data in `ProcessedData/`:

1) inspect the two CSV files in `ProcessedData/`;
2) run the corresponding analysis script, for example `python AnalysisCode/analysis_count.py`;
3) inspect the generated outputs in `Results/`.

Each script performs a separate analysis of scam categories, platforms, items, payments, losses, success rates, or recovery outcomes, with no required execution order.

D. Major Claims

The artifact supports the reproduction of the paper's main empirical findings. Specifically, the study characterizes a diverse second-hand marketplace scam landscape comprising 43 fine-grained categories across 3,392 processed cases. The released structured case representations, category labels, confidence scores, and associated metadata are provided in `ProcessedData/`, while the implementation of the LLM-assisted filtering, structured extraction, and taxonomy construction pipeline is provided in `LLMPrompt/`. Based on the released processed data, the major quantitative results reported in the paper can be reproduced using the corresponding scripts in `AnalysisCode/`, including analyses of scam-category distributions, platform-level differences, scam-related items, payment methods, victim financial losses, reported scam success rates, and recovery attempts and outcomes.

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APPENDIX

A. Categories in the SHScam Taxonomy

We initialize the taxonomy with eight seed categories commonly observed in second-hand marketplace scams, including A3, D1, D2, D3, D4, C1, C2, and E3. Through our iterative classification and refinement process, we identify an additional 35 categories, resulting in a 43-category taxonomy for second-hand marketplace scams. The complete definitions of all 43 scam categories are available in our online appendix: https://sites.google.com/view/ndss-onlineappendix.

B. Fine-Grained Scam Attributes

We identified seventeen fine-grained attributes to characterize scam cases. Extracting these attributes from victim reports enables systematic analysis of scam patterns and behaviors. Figure 12 illustrates how a Reddit scam report is converted into structured JSON using these attributes.

- **victim_role**: Victim's role (buyer/seller/other).
- **scammer_role**: Scammer's role (buyer/seller/other).
- **item_traded**: Traded item or category.
- **item_value**: Listed or estimated item price.
- **has_real_payment**: Whether actual payment occurred.
- **payment_channel**: Payment method (e.g., PayPal, Venmo, bank transfer).
- **transaction_stage**: Stage when the scam occurs (e.g., before payment, after payment, after shipping).
- **trigger_event**: Action initiating the scam.
- **scam_mechanism**: Deceptive techniques (e.g., forged payment proof, off-platform redirection).
- **behavior_sequence**: Chronological sequence of actions taken by the scammer and victim throughout the scam.
- **channel_migration**: Cross-platform movement (from, to, reason), if any.
- **scam_success**: Whether the scam succeeded.
- **victim_loss_amount**: Victim's financial loss (0 if none).
- **scam_targets**: Assets or information targeted by the scammer (e.g., money, personal information, or accounts).
- **recovery_channel**: Channel through which the victim attempts to recover losses (e.g., platform support, payment provider, or law enforcement).
- **recovery_outcome**: Outcome of the victim's recovery attempt (e.g., successful, unsuccessful, or none).
- **platform**: Transaction platform (e.g., Goofish, eBay, Craigslist, Facebook Marketplace).

C. Supplementary Charts

The following supplementary figures are provided for a better understanding of the paper, including Figure 10, Figure 11, and Figure 12.

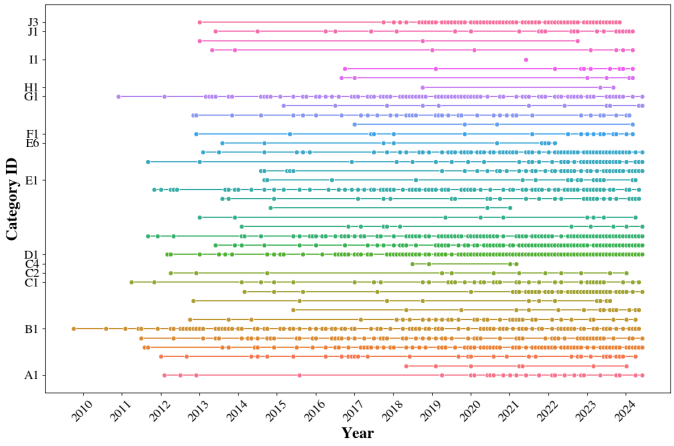


Fig. 10: Time distribution of scams collected from Reddit.



Fig. 11: Specific posting requirements in the proxy-driven scam model, including designated contact information, and an example product listing.


```
{
  "victim_role": "seller",
  "scammer_role": "buyer",
  "item_traded": "bike",
  "item_value": "uncertain",
  "has_real_payment": "no real payment",
  "payment_channel": "Zelle",
  "transaction_stage": "no order & no payment",
  "trigger_event": "Buyer proposed Zelle payment and pickup by brother without cash.",
  "scam_mechanism": [
    "forged payment proof",
    "forged official emails"],
  "behavior_sequence": [
    "Seller lists bike on Facebook Marketplace",
    "Buyer contacts with backstory, proposes Zelle payment and pickup without cash",
    "Seller provides Zelle details",
    "Seller receives forged Zelle email stating funds held due to non-business account",
    "Scammer instructs seller to send money to 'upgrade' the account to release the payment",
    "Seller sends $400 via Zelle",
    "Seller realizes the scam; no buyer payment is received and item is not delivered" ],
  "channel_migration": [
    {
      "from": "Facebook Marketplace",
      "to": "Email",
      "reason": "Process payment and send requested 'upgrade' transfer "
    }],
  "scam_success": "successful",
  "victim_loss_amount": "USD 400",
  "scam_targets": ["money"],
  "recovery_channel" : ["none"],
  "recovery_outcome": "no_attempt",
  "platform": "Facebook Marketplace"
}
```


Fig. 12: Structured representation of the example case.